

Plant Disease Detection using CNN

Problem Statement

Title: Plant Disease Detection using Convolutional Neural Networks

Objective:

To build a deep learning-based image classification model that can automatically identify plant diseases from leaf images. This model will help farmers and agricultural professionals detect diseases early and take appropriate action, improving crop yield and minimizing loss.

Description:

Plant diseases can significantly affect agricultural productivity and food supply. Manual inspection is time-consuming and prone to human error. This project aims to develop a Convolutional Neural Network (CNN) model trained on a dataset of healthy and diseased plant leaves to classify and detect diseases efficiently. The dataset is categorized by plant types and diseases, and is processed through image preprocessing and augmentation before training.

CNN Image Classification Pipeline

1. **Data Collection:** Gather a dataset of plant leaf images labeled with corresponding diseases.
2. **Data Preprocessing:** Resize images, normalize pixel values, and split the data into training, validation, and testing sets.
3. **Data Augmentation:** Apply transformations (rotation, flipping, zooming, etc.) to increase dataset diversity and reduce overfitting.
4. **Model Design:** Build a CNN architecture with convolutional, pooling, and fully connected layers.
5. **Compilation:** Choose an optimizer (e.g., Adam), loss function (e.g., categorical cross-entropy), and evaluation metric (e.g., accuracy).
6. **Training:** Train the CNN model on the training dataset using the specified parameters and monitor performance on the validation set.
7. **Evaluation:** Assess model accuracy and performance using the test dataset. Generate classification reports and confusion matrix.
8. **Deployment:** Export the trained model and integrate it into an application or system for real-time prediction.